

The effect of voucher and public school education on Chilean students' mathematics and reading test score trajectories

Valentina Giaconi, Pascal Bressoux & Patricio Felmer

To cite this article: Valentina Giaconi, Pascal Bressoux & Patricio Felmer (2022) The effect of voucher and public school education on Chilean students' mathematics and reading test score trajectories, School Effectiveness and School Improvement, 33:2, 303-331, DOI: [10.1080/09243453.2021.2014896](https://doi.org/10.1080/09243453.2021.2014896)

To link to this article: <https://doi.org/10.1080/09243453.2021.2014896>



View supplementary material [↗](#)



Published online: 28 Dec 2021.



Submit your article to this journal [↗](#)



Article views: 138



View related articles [↗](#)



View Crossmark data [↗](#)



The effect of voucher and public school education on Chilean students' mathematics and reading test score trajectories

Valentina Giaconi ^a, Pascal Bressoux ^b and Patricio Felmer^c

^aInstituto de Ciencias de la Educación, Universidad de O'Higgins, Rancagua, Chile; ^bLaboratoire de recherche sur les apprentissages en contexte EA602, Université Grenoble Alpes, Grenoble, France; ^cCenter for Mathematical Modeling and Center for Advanced Research in Education, Institute of Education, Universidad de Chile, Santiago, Chile

ABSTRACT

This article aims to measure and compare the effectiveness of voucher schools and public schools on mathematics and reading test score trajectories from Grade 4 to Grade 10 in Chile. We analysed a national database that includes measures taken in 2007, 2011, and 2013 from a same sample of over 140,000 students. Multilevel-growth models were used to analyse the trajectories, and linear adjustment with propensity score matching was used to control for confounding variables. We found positive effects of voucher education on intercepts which, after controlling for schools' socioeconomic composition and selection policies, became irrelevant in terms of effect sizes. Regarding slopes, voucher schools have a significant positive effect, which is irrelevant in terms of effect sizes if 1 year is considered, but relevant for a 12-year school trajectory. The results illustrate a stratification between the public and voucher systems during a period when intense neoliberal policies were enforced.

ARTICLE HISTORY

Received 14 April 2020

Accepted 2 December 2021

KEYWORDS

Public schools; voucher schools; multilevel growth modelling; propensity score matching; student trajectories

Introduction

Voucher schools are defined as schools with private management receiving funding from the public system through student vouchers. They are usually seen as an alternative to public schools. The study of voucher and public educational systems is critical because market-oriented system reforms are supported by the popular idea that private schools are more effective than public schools (Bellei, 2008). Besides, voucher programmes have been implemented in several countries whether at a small or large scale (Epplé et al., 2017; Organisation for Economic Co-operation and Development [OECD], 2012).

The case of Chile is compelling because it is an example of the implementation of intense neoliberal policies at the national level over several decades. Chile is the country with the largest private voucher system (Bettinger, 2011). Currently, 49.7% of schools are part of the voucher system, while 44.8% of schools are public and 5% are private (Ministerio de Educación, 2014). Until 2017, voucher and public schools

constituted two different school systems, with different rules that regulated their operation and dynamics (Bellei, 2018).

Moreover, there are other specific characteristics of the Chilean voucher system that make it an important case to study. First, Chile has a universal voucher system, which means that vouchers are open to all students. This is different from countries with targeted voucher systems that favour disadvantaged students. An OECD study (2012) found that the systems with universal vouchers had twice as much stratification as systems with targeted vouchers or no vouchers at all. Second, until the year 2016, Chile allowed voucher schools to select students and charge family fees; however, public schools could only select students if they had more applicants than places and cannot charge family fees. The OECD found that allowing schools to select students increases social stratification.

In summary, Chile is an important case study because of the extent and the duration of its voucher system, the universality of voucher access, and the license of schools to select students. These characteristics were all present during the years studied in this article, although they have now changed following the implementation of a new law for public education which came into force in 2016 (Law N° 20845, 2015).

Several studies compare private and public education with Chilean data. These studies present mixed conclusions and use different methodological strategies (Bellei, 2008). The most important limitation in all these studies is selection bias (Bellei, 2008). To estimate the true effect of schools, it is necessary to control for several factors such as socioeconomic and academic segregation, peer effects, the selection of students by schools, and the selection of schools by families (Bellei, 2008). Selection bias creates a significant obstacle to identifying the effectiveness of a school because students are not assigned randomly to their schools (Raudenbush & Willms, 1995). Therefore, there are many methodological issues to resolve. A typical strategy for addressing this problem is to control for confounding variables, which can be done using different statistical methods, for example, regression analysis or propensity score analysis.

Another critical issue when identifying the effectiveness of a school is the study of student trajectories. Analysing academic trajectories improves the estimation process because it accounts for measurement error and allows us to study the shape of the change and the rate of the growth (Bressoux, 2010). Guldmond and Bosker (2009) stated that studying trajectories increased the likelihood of detecting school effects because it is a more reliable measure of student progress.

This article aims to estimate and compare the effect of voucher schools and public schools on student trajectories from Grade 4 to Grade 10 in mathematics and reading.

With the available data, which cover trajectories of 6 years, we are interested in estimating the effect of being in the voucher-school system compared to being in the public-school system. This study extends previous research performed with Chilean data in four ways: (a) It investigates students' trajectories; (b) it considers a longer period that includes primary and secondary education; (c) it uses a census-based sample; and (d) it focuses specifically on the selection bias problem between voucher and public schools.

Theoretical background

Voucher and public schools: international context

The common characteristic of voucher programmes is that they are a public subsidy to private schooling intended to provide family selection options, with the final objective of increasing school competition (Morgan et al., 2015). To increase school competition, usually voucher schools have greater autonomy (Morgan et al., 2015; Zimmer et al., 2019). On the one hand, the theory that favours voucher programmes argues that vouchers will increase the (a) quality of education because schools need to compete for students; (b) equity because students will have access to private schooling; (c) educational variety because of greater autonomy of voucher schools; and (d) general school system quality because public schools also need to compete with voucher schools (Epple et al., 2017; Morgan et al., 2015). On the other hand, the theory opposing vouchers proposes that vouchers will produce “cream skimming” processes, where voucher schools will eventually absorb the more advantaged students from public schools. This will increase inequality and stratification through peer effects and the possible stigmatization of students. The stratification will concentrate disadvantaged students in public schools, increasing the costs of these schools. Finally, positive effects of voucher schools are based on the ability of families to choose, which is higher in families with more social capital (Epple et al., 2017; Morgan et al., 2015). These theories should be analysed in light of the empirical differences of voucher programmes like: (a) the size of the programme, for example, the small-scale programmes implemented in the US or large-scale voucher programmes like Chile and Sweden (Epple et al., 2017); (b) universal vouchers provided to all the students (e.g., Chile) or targeted vouchers (e.g., Colombia); (c) flat vouchers or differential amounts of voucher for different groups; (d) whether or not school selection of students is permitted; (e) whether or not for-profit schools are permitted; and so forth. These differences can increase the positive or negative outcomes of a voucher system, and imply that conclusions regarding the effects of voucher systems should be drawn in relation to the specific implementation and context.

Research into the effect of vouchers has been undertaken in several reviews regarding experiences around the world (Epple et al., 2017), both in the US (Betts & Tang, 2019; Zimmer et al., 2019) and in developing countries (Morgan et al., 2015). Several effects of voucher schools can be studied, such as the effects of voucher schools on achievement, school attendance, access to college, or the performance of the whole school system across several indicators (Zimmer et al., 2019). In this study, we focus on the effect of voucher schools on achievement, and we summarize some relevant results regarding this.

Regarding developing countries, Morgan et al. (2015) concluded that there was little rigorous evidence on this type of country, and analysed the cases of Colombia, Pakistan, and Chile. In Colombia, they analysed the PACES programme (Programa de Ampliación de Cobertura de la Educación Secundaria [Programme for expanding the coverage of secondary education]), which is a targeted voucher system that provides vouchers to lower income students and allows them to attend secondary private schools in highly populated low-income areas. In some cases, the voucher did not cover the total cost of the private school. Students were assigned by lottery, and, in order to remain in the programme, they could not repeat a grade. There are reports of positive effects on

achievement (Bettinger, 2011; Morgan et al., 2015). Also, Bettinger et al. (2019) found positive long-term effects on tertiary education and labour-market access, specifically on the students who attended vocational secondary private schools. In Pakistan, the voucher programme helped to increase access to education, especially to girls (Morgan et al., 2015). With respect to Chile, the results will be discussed in the next section.

Another important and complex example is the USA, where voucher and charter school systems have been installed in several states. Charter schools are privately managed and receive public funding, although not necessarily in the form of vouchers. Regarding the effectiveness of voucher and charter schools, studies show mixed conclusions regarding achievement, except for lottery-based designs, where there is more evidence favouring charter schools; however, this evidence is limited because this design can only evaluate oversubscribed charter schools (Zimmer et al., 2019). It is important to mention that in non-achievement outcomes there is more conclusive evidence favouring charter schools (Zimmer et al., 2019). Betts and Tang's (2019) meta-analysis showed small positive effects of voucher schools that are meaningful considering the cumulative effect during several school years. They also found, however, a lot of geographical variation, showing that charter schools have better results in some specific areas.

In Europe, there are more examples where voucher systems have been implemented and studied. For example, in Spain, studies have found small positive effects of voucher schools; however, they differ between regions (Crespo-Cebada et al., 2014). With respect to the Swedish voucher system (independent schools), Bettinger (2011) concludes that there is a small positive effect on average grades and Böhlmark and Lindahl (2015) found positive effects on long-term educational outcomes.

As a conclusion, it is not possible to claim that in general the voucher school system has either a positive or negative effect on student achievements, because the implementation of such systems differs between countries and regions, and each system has a different underlying hypothesis (Bettinger, 2011). However, in some cases with specific system needs and designs, voucher schools show an improvement on achievement or long-term results (e.g. Colombia, Pakistan, Sweden, and some cases in the USA).

Another issue is the methodological design to evaluate the effect of the vouchers. Lottery-based designs are preferred because they rule out non-observed characteristics of the students that may affect the results; however, they only permit the evaluation of oversubscribed voucher schools (Zimmer et al., 2019). Observational studies that try to equate control and treatment groups controlled by previous achievement and confounding factors (as linear regression and propensity score matching) have the issue of being affected by non-observable characteristics of students. However, studies comparing lottery-based and observational studies have arrived at the same conclusions (Zimmer et al., 2019), suggesting that observational studies can be informative.

Also, in the study of the efficacy of voucher and private schools it is essential to understand the underlying mechanism that makes voucher schools better or worse (Bettinger, 2011) in order to get useful results that can inform policy debates. Here, it is useful to consider Raudenbush and Willms' (1995) Type A and Type B school effects. Type A effects consider the effect of the school without distinguishing whether it arises from the school context (e.g., being in a privileged neighbourhood or having students with good previous achievement) or from good school practices (e.g., having efficient staff). Type A effects are

important for parents, who want their kids to have good results regardless of where those results come from. Type B effects control for the school context in order to isolate school practice efficacy, so they are more important for policymakers (Raudenbush & Willms, 1995).

This study focuses on the effect of voucher schools on students' long-term achievement trajectories during school, modelling not only a final outcome but the effect of the schools on the shape of the student's trajectory. There are studies that report on the long-term effects of vouchers (e.g., Angrist et al., 2006; Bettinger et al., 2019; Böhlmark & Lindahl, 2015). However, these studies focused on the effect of voucher schools on final outcomes, usually after schooling ends, not on the effect on school trajectories. Also, having more than two measures allows for the distinction of variance between and within subjects. A second contribution of this study is that it is placed in a country with a large-scale and long-term voucher school policy. In this sense, this study provides new insights regarding the voucher systems.

Voucher and public schools: Chilean context

The voucher system in Chile was implemented in 1980. The educational system design generated three types of schools: public schools funded and managed by public entities; voucher schools managed by private entities and funded by the state; and private schools funded and managed by private entities. Since the creation of voucher schools, the public sector has been shrinking systematically (Hofflinger & von Hippel, 2020), and now the voucher part of the system educates the majority of students. During the years analysed in this study (2007–2013), the characteristics of the voucher system were as follows: (a) It was a universal voucher programme, meaning all students can access it; (b) the voucher amount was flat, that is, the same for all students; (c) voucher schools were allowed to charge fees to the families while public schools were free; (d) voucher schools were allowed to be for-profit schools; and (e) voucher schools were allowed to select students, while public schools only could select if they had more applicants than vacancies. Characteristics (c), (d), and (e) were changed in 2017 (Bellei, 2018). Characteristic (b) was changed in 2008, where a targeted voucher programme was implemented to increase the voucher for students with lower socioeconomic conditions. The target-voucher law (Ley SEP) was enacted in 2008, allowing both public and voucher schools to use this targeted voucher if they have students in the low-income group defined by the law. This law was implemented progressively, and it was not fully enacted until 2013 (Valenzuela et al., 2013); many studies have evaluated its effectiveness (Correa et al., 2014; Mizala & Torche, 2017; Raczynski et al., 2013).

According to the characteristics (a), (b), and (e) of the Chilean school system during the years under study, it is expected to find important levels of stratification, since the OECD (2012, p. 36) indicates that these characteristics are specially associated with more stratified systems. In fact, rough comparisons of student achievement have shown systematically that public schools achieve lower results than voucher schools. However, these differences cannot be attributed to the efficacy of each type of school, because of the enormous disparity in student background characteristics. Chile has a very large between-school variance (Manzi et al., 2014) and a high level of stratification between public, voucher, and private schools. This can be partially explained by the high level

of socioeconomic segregation between schools (Valenzuela et al., 2014) and the autonomy that private and voucher school principals have regarding decisions such as student selection, curricular content, student policy, and human and financial resources (Weinstein & Muñoz, 2014).

Studies regarding the effect of public and private education using Chilean data show mixed results in terms of which type of school leads to better learning gains. In his review, Bellei (2008) shows that even studies using the same data set can draw different conclusions. He identified several issues regarding the methodology that is used to compare public and private schools, concluding that the most significant problem is the high selection bias between these two types of educational alternatives, and identifying the following confounding factors:

- Differing socioeconomic composition: This can have an effect at the student level, and a compositional effect at the class, school, or local authority levels. Moreover, it is necessary to distinguish between structural or economic variables and cultural or social variables.
- Differing academic composition: This can have an effect at the student level, and a compositional effect at the class or school level.
- School selection by families.
- Student selection by schools.

After Bellei's review (2008), new studies have compared public and voucher schools with novel methodological and statistical techniques. For example, Anand et al. (2009) compared public schools, free voucher schools, and voucher schools that charge a family fee. They found that voucher schools that charge a family fee had test-score gains of 0.2 standard deviation over public schools. Nevertheless, the authors state that this identification strategy is limited because it adjusts only for observed characteristics.

Another example is the study of Lara et al. (2011), who used a strategy based on propensity scores and a changes-in-changes estimation method. They exploited the fact that many public schools have only primary education (until Grade 8), so their students have to change schools to continue into secondary education. With a sample constituted by all the students who attended public schools until Grade 8, they compared national test scores in Grade 10 from students who continued to attend public education with the scores from the students who switched to voucher schools. They found small effects in favour of voucher schools that were about 0.04 to 0.06 of a standard deviation.

Manzi et al. (2014) estimated the value added by schools using panel data with two SIMCE (Sistema de Medición de la Calidad de la Educación [Education Quality Measurement System]) national test measures. They used the educational level of parents as instrumental variables to address endogeneity problems. They found that the effectiveness of public and voucher schools changed when they used the instrumental variables. Finally, Zubizarreta and Keele (2017) compared voucher and public schools using a novel matching strategy that accounts for the clustered nature of the measures. They matched students and schools and then compared scores in standardized achievement tests. Their findings show non-significant differences between voucher and public schools and an effect size of 0.027 standard deviation.

In conclusion, studies that compared public and voucher schools in Chile show that differences are negligible or slightly favour voucher schools. Nevertheless, comparing the effect of these types of schools on student trajectories is an ongoing problem, which has only been addressed in primary grades by the work of Ortega Ferrand (2016, pp. 185–187), who found no effects on student growth. This interesting study did not address selection bias problems between voucher and public schools and did not consider measures at the secondary education level, so that an extension would be much valuable.

Methodology

Sample

The sample was taken from the census-based assessment implemented by the System of Measurement of the Quality of Education (SIMCE). The sample comprises SIMCE assessments from years 2007, 2011, and 2013, in Grades 4, 8, and 10, respectively. This longitudinal sample represents Chilean students who attended public and voucher schools and who were on track between Grade 4 and Grade 10. We did not include students attending private schools because they represent only a small part of the population and are a group with high socioeconomic status. Therefore, they come from a context that is very different from that of students attending vouchers and public schools (for further details, see the comparison between students from public, voucher, and private schools in Appendix 1).

We selected students with a valid sample identifier for 2007, 2011, and 2013, and who attended public or voucher schools during these years. Within these conditions, the total sample includes 141,584 students (Table 1).

For the estimation of the initial unconditional multilevel growth model, we used the total sample of 141,584 students (Table 1). This model is useful for establishing the structure of the school effects, considering the longitudinal structure of the data and student mobility between schools.

For the estimation of public versus voucher school effects on student trajectories, we decided to exclude from the sample students who switched from the voucher system to the public system (and vice versa) during the investigated period. This was done because they represent specific situations that are difficult to generalize and the sample sizes for

Table 1. Sample sizes for each statistical analysis.

Objective	Analysis	Sample size
Determination of the structure of school effects	Unconditional multilevel growth models	141,584 (39,699 students that attended only public schools, 57,275 students that attended only voucher schools, and 44,610 students that switched between systems)
Estimation of the effectiveness of public schools over voucher schools	Multilevel growth models with linear adjustment of confounding variables	96,974 (39,699 students in public schools and 57,275 students in voucher schools)
	Multilevel growth models with linear adjustment of confounding variables and propensity score matching.	44,684 (matched sample determined using propensity score matching, 22,342 students in public schools and 22,342 students in voucher schools)

the different switching combinations are too small to guarantee enough common support for comparisons. Therefore, only those students who were enrolled in the same school system at the three measurements were retained, which produced a final sample of 96,974 students (Table 1). Using this sample, we estimated the effect of voucher schools versus public schools using three analytic strategies: linear adjusting, propensity score matching, and both strategies simultaneously. For linear adjusting, the total sample size was used (96,974). Propensity score matching produces a sample reduction to the group of students with common support, which allows matching between the treatment and control conditions. The sample size for the analysis done in the matched sample was 44,684 students (Table 1).

Variables

In this subsection, we describe each of the variables considered in this study, according to their role in the analysis:

- **Treatment:** The treatment variable *SchoolSystem* defines possible treatments. It is a categorical variable. Category 1 corresponds to students who attended voucher schools in Grades 4, 8, and 10. Category 0 corresponds to students who attended public schools in Grades 4, 8, and 10.
- **Outcome variables (student achievement measures):** The outcome variables *normalized mathematics score* and *normalized reading score* are normalized scores in mathematics and reading from the national standardized tests implemented by SIMCE. The tests aim to evaluate student learning in different subjects. For each subject, they are designed to evaluate the contents and abilities specified in the national curriculum (Agencia de Calidad de la Educación, 2015), and are applied close to the end of the school year. The SIMCE test scores are not designed to be compared longitudinally between different levels, which means it is not correct to model trajectories on the test score scale given by SIMCE. To make scores comparable, we normalized the SIMCE test scores before merging the databases. Therefore, our analyses will not show absolute progress, but will instead show differential progress relative to the comparison of groups of students, knowing that the average progress for the total sample is, by construction, null.
- **Control variables (student socioeconomic and expectations measures):** The control variables corresponding to *mother's education*, *father's education*, *number of books in the house*, *computer*, *internet*, *income*, *parents' expectations*, and *attendance to pre-school* came from the parents' questionnaire. This questionnaire is applied to the student's parents or guardians during the days of application of the SIMCE test (Agencia de Calidad de la Educación, 2015, p. 59). The variables *mother's education*, *father's education*, *income*, and *parents' expectations* were transformed into numeric variables according to the conversion tables from (Agencia de Calidad de la Educación, n.d.). Likewise, we re-coded the variable *number of books in the house*. After this procedure, all the variables had numeric values. The variables *computer*, *internet*, *Kindergarten*, *Pre-kindergarten*, *ECE 0–2* (*early childhood education for children of 0 to 2 years of age*), *ECE 2–4* (*early childhood education for children of 2 to 4 years of age*) are dummy variables. In the case of *computer* and *internet*, the value is 1 if the student has a computer or internet connection at home and 0 if not. The variables *Kindergarten*, *Pre-kindergarten*, *ECE*

0–2, and ECE 2–4 have the value 1 if the student has attended the corresponding level of preschool education and 0 if not.

- Treatment variation (school aggregated variables): In Chile, school socioeconomic segregation is extreme, which implies that school composition can vary massively. Therefore, it is imperative to consider compositional variables at the school level estimated using aggregated student-level variables. The student-level variables used to define school aggregate variables are socioeconomic indicators, attendance to pre-school, and school selection (Table 2). The school selection variables came from the parent questionnaire. They correspond to questions regarding what the school required before accepting the student. Examples of this include questions asking if the school applied a test and required an income certificate. To develop the school-level indexes, we calculated the school average for each variable (for all the compositional variables names, we added an “sc” in superscript to denote that they variate at the school level). In order to ensure each variable was treated equally, we standardized each of them. To develop the indexes, we carried out principal component analyses, the results of which are summarized in Table 2.

Data analysis

For the modelling of trajectories, we used multilevel growth models, which are also flexible enough for modelling students’ mobility between schools through cross-classified structures. This is especially important during the transition between primary and secondary education, where it is more common to see students change schools, because there are schools that only have primary levels.

Two methodological problems appeared in this study: selection bias and missing data. The main problem in comparing public versus voucher schools in Chile is selection bias (Bellei, 2008). All three of our modelling strategies to adjust for selection bias (linear adjustment for confounding variables, propensity score matching, and both methods simultaneously – where we used the matched data and added the confounding variables as covariates) were implemented in conjunction with multilevel models, to allow us to measure the treatment effect on the random slopes and random intercepts and also to be able to include school aggregated variables. The strategies are presented in Table 3.

Table 2. School indexes definition.

School Index	Variables	Interpretation
SES	Mother’s education ^{sc} Father’s education ^{sc} Number of books ^{sc} Home income ^{sc} Computer ^{sc} Internet ^{sc}	Schools with higher values in this index are composed of students with better socioeconomic conditions
SES Selection	Marriage certificate ^{sc} Church-related certificate ^{sc} Income Certificate ^{sc} Game session ^{sc}	Schools with higher values in this index asked more frequently for socioeconomic-related certificates before accepting a student
Academic Selection	Preschool certificate ^{sc} Grades certificate ^{sc} Test ^{sc}	Schools with higher values in this index asked more frequently for academic-related certificates or tests before accepting a student
Preschool attendance	Kindergarten ^{sc} Pre-kindergarten ^{sc} ECE 0–2 years ^{sc} ECE 2–4 years ^{sc}	Schools with higher values in this index are composed of students with more years of pre-school education

Note: SES = socioeconomic status; sc = school-level variable; ECE = early childhood education.

Table 3. Models for treatment effect estimation.

Model	Inclusion of control variables	Inclusion of school aggregated variables	Type of effect
1	None	No	–
2	Linear adjusting	No	A
3	Propensity score matching	No	A
4	Linear adjusting and Propensity score matching	No	A
5	Linear adjusting	Linear adjusting	B
6	Propensity score matching	Linear adjusting	B
7	Linear adjusting and Propensity score matching	Linear adjusting	B

Regarding the missing data issue, the percentage of missing data for each variable ranges between 0% and 48%, with most of the variables missing 25% (for more details in the percentages of missing data, see Appendix 2). This issue was addressed through multiple imputation techniques. The imputation models have to preserve the relevant characteristics of the data (Enders et al., 2016). In our case, the hierarchical nature of the data structure is the key aspect to be preserved. The intra-class correlations of most of the variables, using the grouping of schools in Grades 4, 8, and 10, were very high. Therefore, we used imputation techniques specific to multilevel data. We estimated 20 imputations.

The combined use of these methods was achieved with the following strategies:

- All the estimated treatment effects were obtained using multilevel models. Multiple imputation techniques are compatible with the estimation of multilevel models.
- Linear adjusting was implemented including covariates (control and school-level variables from Table 3) in the multilevel models.
- To implement the propensity score matching, first, we fitted a logistic regression that predicted, for each student, the probability of being in a public school. We used all the control variables as covariates, at the occasion and the student levels described in Table 3. Also, we added all the potential interactions between two variables and all the quadratic terms for the numerical variables. We used non-parsimonious models because the focus for propensity score modelling is to achieve balance more than parsimony. For each of the imputed data sets, we estimated the propensity score model and extracted the estimated propensity scores. Then, for each observation, we averaged the 20 propensity scores, getting only one propensity score for each. With these estimated propensity scores, we matched students from public schools with students from voucher schools. The matching method was nearest neighbourhood matching with a caliper equal to 0.2 times the standard deviation of the logit of the propensity scores (caliper = 0.26). This caliper was chosen according to recommendations from Austin (2011). To evaluate covariate balance, we compared the standardized mean difference (SMD) between students in public schools with students in voucher schools in the complete and the matched sample (Pattanayak, 2015). We did not consider the significance of the mean differences because it is not recommended, as it depends on sample sizes that vary for different matching strategies (Pattanayak, 2015).

All the estimation processes were done using the software R. Propensity score matching was done with the Matching package (Sekhon, 2011). For multilevel models, we used the

package lme4 (Bates et al., 2015) and, for multiple imputation, the Mice (van Buuren, 2011) and Pan (Zhao & Schafer, 2016) packages. Multiple imputation procedures and the estimation of multilevel models with cross-classified random effects was very demanding in terms of computational resources. To deal with this problem, we used the super-computing infrastructure of the National Laboratory for High Performance Computing NLHPC (ECM-02).

Modelling strategies

The first modelling step was to define the multilevel growth models using a stepwise strategy (Singer & Willett, 2003, pp. 92–101). We used the same model for reading and mathematics test scores. For the modelling of *time*, we defined *time* = 0 for the measures taken in Grade 10. Therefore, the intercepts of the growth curves correspond to the student test scores at Grade 10. The values for *time* for the measures at Grades 4 and 8 were assigned in order to have a variance equal to one (*time* = –2,405 at Grade 4, *time* = –0,802 at Grade 8). We restricted the variance of the *time* variable to avoid estimation problems in the multilevel models.

The second modelling step was controlling for selection bias using linear adjusting and propensity score matching. For this, we distinguished between the student-level variables and the school-level variables. We considered the school-level variables as treatment variations because they vary at the school level and also depend on school-level decisions. We tested models with and without school aggregated variables. These options allow us to estimate Type A and Type B school effects according to Raudenbush and Willms' (1995) classification.

In our study, when we controlled only for student-level variables, our estimation of the voucher versus public school effects corresponds to a Type A effect (Models 2, 3, and 4, Table 3). When we controlled for student-level variables, school compositional effects, and selection practices, our estimation of the voucher versus public school effects corresponds to a Type B effect (Models 5, 6, and 7, Table 3).

Results

Background characteristics of the treatment groups

We estimated the mean and standard deviation in each control variable for the students in voucher and public education. In addition, we presented standardized mean differences (SMD) interpreted using guidelines from Funder and Ozer (2019).

It is clear from Table 4 that in the complete sample there are differences between students attending public and voucher schools, with clear disadvantages for students in public schools. In terms of the differences in access to learning resources such as the internet, computers, and books, their size varies from medium to large, and slightly decreases over time, especially the access to a computer. The differences regarding socioeconomic conditions such as parents' education and income are large across all the measures. For the other variables, the differences are sizable. For attendance of pre-school, the standardized mean differences are medium to small.

In summary, there is a high amount of selection bias that remains stable during primary and secondary education.

Table 4. Descriptive statistics and standardized mean differences (SMD) for background variables.

Variable	Complete sample			Matched sample		
	Public Mean (SD) N = 39,699	Voucher Mean (SD) N = 57,257	SMD	Public Mean (SD) N = 22,342	Voucher Mean (SD) N = 22,342	SMD
Computer (Grade 4)	0.32 (0.47)	0.66 (0.48)	−0.73	0.47 (0.49)	0.42 (0.49)	0.11
Computer (Grade 8)	0.75 (0.42)	0.9 (0.31)	−0.42	0.85 (0.36)	0.81 (0.39)	0.11
Computer (Grade 10)	0.81 (0.37)	0.93 (0.28)	−0.36	0.88 (0.32)	0.85 (0.34)	0.07
Internet (Grade 4)	0.09 (0.31)	0.35 (0.46)	−0.63	0.16 (0.36)	0.12 (0.33)	0.08
Internet (Grade 8)	0.46 (0.49)	0.74 (0.44)	−0.59	0.6 (0.48)	0.54 (0.49)	0.12
Internet (Grade 10)	0.53 (0.48)	0.79 (0.42)	−0.59	0.65 (0.46)	0.62 (0.47)	0.08
Father's Education (Grade 4)	9.67 (3.32)	12.56 (3.26)	−0.88	11 (3)	10.5 (3.01)	0.15
Father's Education (Grade 8)	9.64 (3.36)	12.52 (3.34)	−0.86	10.96 (3.06)	10.41 (3.08)	0.16
Father's Education (Grade 10)	9.61 (3.39)	12.5 (3.4)	−0.85	10.92 (3.13)	10.39 (3.14)	0.16
Mother's Education (Grade 4)	9.69 (3.27)	12.57 (3.13)	−0.9	11.1 (2.9)	10.57 (2.93)	0.17
Mother's Education (Grade 8)	9.71 (3.31)	12.56 (3.19)	−0.88	11.09 (2.94)	10.52 (2.98)	0.18
Mother's Education (Grade 10)	9.71 (3.32)	12.56 (3.22)	−0.87	11.06 (3)	10.54 (3.04)	0.16
Number of books (Grade 4)	19.76 (26.36)	35.06 (33)	−0.5	26.27 (28.92)	23.79 (27.59)	0.08
Number of books (Grade 8)	29.74 (28.96)	45.46 (35.36)	−0.48	36.43 (31.36)	33.74 (29.85)	0.08
Number of books (Grade 10)	32.15 (30.63)	48.67 (36.3)	−0.48	38.93 (32.5)	36.29 (31.19)	0.08
Income (Grade 4)	1.88 (1.78)	3.73 (3.08)	−0.7	2.38 (1.92)	2.16 (1.7)	0.08
Income (Grade 8)	2.23 (2)	4.47 (3.96)	−0.68	2.8 (2.25)	2.49 (1.83)	0.09
Income (Grade 10)	2.73 (2.5)	5.21 (4.45)	−0.66	3.37 (2.7)	3.09 (2.34)	0.08
Parents' expectations (Grade 4)	14.47 (2.79)	16.41 (2.19)	−0.79	15.53 (2.41)	15.29 (2.51)	0.1
Parents' expectations (Grade 8)	15.37 (2.3)	16.72 (1.75)	−0.68	15.95 (2.05)	16 (2)	−0.02
Parents' expectations (Grade 10)	15.77 (1.99)	16.94 (1.55)	−0.67	16.24 (1.78)	16.34 (1.71)	−0.06
Gender (fraction of girls)	0.51 (0.5)	0.53 (0.5)	−0.03	0.51 (0.5)	0.51 (0.5)	0
ECE (0–2 years)	0.31 (0.45)	0.43 (0.48)	−0.26	0.36 (0.47)	0.33 (0.46)	0.05
ECE (2–4 years)	0.66 (0.44)	0.82 (0.39)	−0.37	0.73 (0.42)	0.74 (0.42)	−0.03
Pre-kinder	0.47 (0.49)	0.63 (0.49)	−0.33	0.53 (0.49)	0.51 (0.49)	0.04
Kinder	0.05 (0.21)	0.09 (0.29)	−0.17	0.06 (0.24)	0.05 (0.23)	0.03

Note: We compare students attending public and voucher education. All the differences were statistically significant considering a significance level of 5%. Means are the pooled estimates from the 20 imputed data bases. ECE = early childhood education.

Regarding the differences between schools, Table 5 illustrates that, for the complete sample, the composition and selection practices of schools generate large differences. Students from voucher schools have peers with a different social composition to the peers they would have at a public school. The effect sizes for these differences are 1.49 and 1.57, which is about twice the size of a large effect. Regarding the attendance to pre-school indexes, the differences are also large. Standardized differences regarding the academic selection and the socioeconomic selection indexes are between 0.63 and 1.66. These results reflect the nature of the system, where voucher schools are allowed to select students, but public schools are not, except in the case that they receive more applicants than there are places available.

Application of propensity score matching

The propensity score was defined as the probability to attend a public school. The estimated propensity scores correlated negatively with the socioeconomic variables, which means that the probability of attending a public school was higher for students with lower socioeconomic conditions. The standardized mean differences between students in the voucher and public system are detailed in Table 4. For all the variables, after matching the standardized mean differences were lower than 0.2. Therefore, we considered that the balance achieved was adequate.

Table 5. Descriptive statistics and standardized mean differences (SMD) for school indexes.

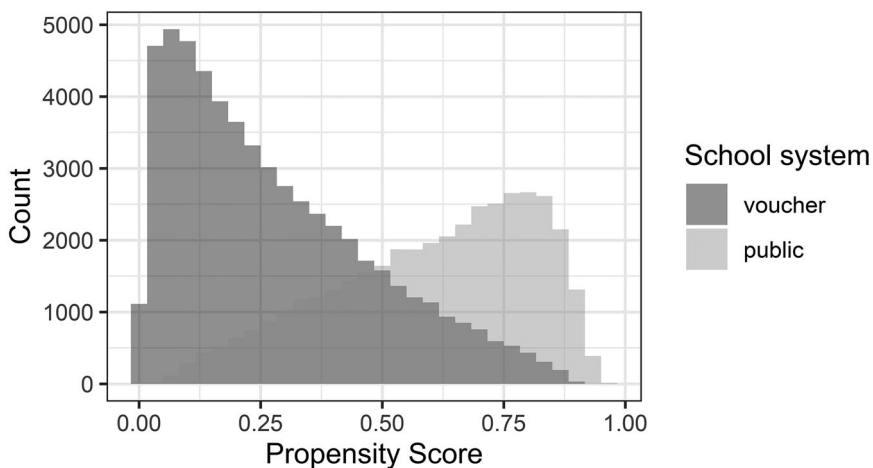
Variable	Complete sample			Matched sample		
	Public Mean (SD) N = 39,699	Voucher Mean (SD) N = 57,257	SMD	Public Mean (SD) N = 22,342	Voucher Mean (SD) N = 22,342	SMD
SES index (Grade 8)	-0.31 (0.54)	0.64 (0.64)	-1.57	-0.09 (0.5)	0.23 (0.6)	-0.52
SES index (Grade 10)	-0.72 (0.55)	0.18 (0.64)	-1.49	-0.52 (0.52)	-0.21 (0.63)	-0.5
Preschool attendance index (Grade 8)	-0.15 (0.75)	0.33 (0.64)	-0.7	0.02 (0.65)	0.1 (0.65)	-0.12
Preschool attendance index (Grade 10)	-0.54 (0.78)	0.08 (0.67)	-0.88	-0.34 (0.67)	-0.18 (0.69)	-0.23
SES selection index (Grade 8)	-0.31 (0.28)	0.68 (1.45)	-0.89	-0.3 (0.26)	0.35 (1.18)	-0.59
SES selection index (Grade 10)	-0.53 (0.2)	0.28 (1.1)	-0.95	-0.52 (0.2)	0.03 (0.9)	-0.64
Academic selection index (Grade 8)	-0.38 (0.65)	0.87 (0.82)	-1.66	-0.22 (0.66)	0.56 (0.81)	-1.03
Academic selection index (Grade 10)	-0.17 (0.9)	0.37 (0.84)	-0.63	-0.07 (0.91)	0.17 (0.85)	-0.28

Note: We compare students attending public and voucher education. All the differences were statistically significant considering a significance level of 5%. Means are the pooled estimates from the 20 imputed data bases. SES = socioeconomic status.

The original sample has 39,699 students from public schools and 57,257 students from private schools. The matched sample has 22,342 students in each group, which implies a significant reduction. Considering the high amount of selection bias, losing so many students is not surprising. Also, the sample reduction implies that without matching we are comparing several units without similar counterparts, and that after matching the studied population changed. In Table 4, we can see that the characteristics of the original sample and the sample after matching differ. The estimated propensity score for each group has been plotted in Figure 1. We can see that there is a large number of students in both extremes of the propensity score distribution who do not have matches. This pattern explains the decrease in sample size in the matched data.

Multilevel growth models for student trajectories

In order to define the multilevel growth model for student trajectories, we estimated unconditional mean models and unconditional growth models in a stepwise manner (Singer & Willett, 2003, pp. 92–101). In the first step, we included a linear effect of time

**Figure 1.** Common support of the estimated propensity scores.

and random intercepts and slopes at the student level. In the second step, we included school effects on the intercepts and slopes, testing all possible combinations of cross-classified school effects on the student intercepts. We selected the best model considering fit indexes (Deviance, Akaike information criterion [AIC], and Bayesian information criterion [BIC]). The results for reading and mathematics are presented in Table 6. These results show that although the residual variance (intra-individual) is lower than the variance at the student level, it remains very important. For both disciplines, it represents about a third of the total variance. It is important to define random effects at the school level, because in all the models, the percentage of variance explained at the school level is important. Also, the percentage of variance explained from the school in Grade 10 is bigger than the variance from the school in Grade 8. When we added random slopes at the school level, the variances of the slopes at the student level decreased. The slopes' variances are small, but it is important to note that the size of these variances depends on how the time is coded. In fact, the difference in deviance between models shows that including random slopes at the school level is relevant. In summary, the school effects on student trajectories are significant.

The final unconditional model for both areas (mathematics and reading) was the following:

$$\begin{aligned} y_{ti(jk)} &= \beta_{1i(jk)} \text{time} + \phi_{i(jk)} + \varepsilon_{ti(jk)} \\ \phi_{i(jk)} &= \beta_0 + \mu_j^{8th} + \mu_k^{10th} + \xi_{i(jk)} \\ \beta_{1i(jk)} &= \beta_{10} + v_j^{8th} + v_k^{10th} + \omega_{i(jk)} \end{aligned}$$

In this model, the outcome variable $y_{ti(jk)}$ is the normalized test score at time t of student i who in Grade 8 attended school j and in Grade 10 attended school k . The term $\phi_{i(jk)}$ corresponds to the student intercept. The intercepts $\phi_{i(jk)}$ correspond to student scores at Grade 10. The term $\beta_{1i(jk)}$ corresponds to the student slope or rate of the growth in the scores. The model includes cross-classified random effects from the school at Grade 8 and Grade 10 for the intercepts (μ_j^{8th} and μ_k^{10th} , respectively) and the slopes (v_j^{8th} and v_k^{10th} , respectively).

The models do not include cross-classified school effects from the schools at Grade 4 because of estimation problems. These problems can be attributed to an identification issue based on the structure of the Chilean educational system. In Chile, primary education ends at Grade 8, and most of the students do not change school during their primary years. Therefore, we cannot distinguish the effects of the schools at Grade 4 and Grade 8. As such, the final unconditional mean model only includes the school effects at Grades 8 and 10.

Estimation of the effects of public schools on student trajectories using different modelling strategies

The effects of public schools on student trajectories were estimated using the following model (Model 1, Table 3) that includes the variable *SchoolSystem*:

$$\begin{aligned} y_{ti(jk)} &= \beta_{1i(jk)} \text{time} + \phi_{i(jk)} + \varepsilon_{ti(jk)} \\ \phi_{i(jk)} &= \beta_0 + \beta_{int} \text{SchoolSystem} + \mu_j^{8th} + \mu_k^{10th} + \xi_{i(jk)} \\ \beta_{1i(jk)} &= \beta_{10} + \beta_{slope} \text{SchoolSystem} + v_j^{8th} + v_k^{10th} + \omega_{i(jk)} \end{aligned}$$

Table 6. Fixed and random effects of the unconditional multilevel growth models.

Parameters	Mathematics				Reading			
	Null	School intercepts	Null growth	Final unconditional model	Null	School intercepts	Null growth	Final unconditional model
Fixed effects								
Intercept (<i>SE</i>)	0 (0.00)	−0.11 (0.01)	−0.106 (0.011)	−0.107 (0.013)	0 (0.00)	−0.06 (0.01)	−0.064 (0.01)	−0.054 (0.011)
Time (<i>SE</i>)			0 (0.001)	0.001 (0.003)			0 (0.001)	0.01 (0.003)
Random intercepts								
Residual	0.31	0.31	0.246	0.246	0.36	0.36	0.316	0.316
Student	0.69	0.37	0.379	0.365	0.64	0.41	0.425	0.414
School (Grade 8)		0.05	0.048	0.047		0.03	0.034	0.032
School (Grade 10)		0.24	0.262	0.34		0.2	0.205	0.249
Random slopes								
Student-time			0.04	0.022			0.027	0.014
Student time-intercepts			0.013	−0.002			0.017	0.005
School (Grade 8) time				0.007				0.005
School (Grade 8) time-intercepts				0.002				0.001
School (Grade 10) time				0.013				0.009
School (Grade 10) time-intercepts				0.053				0.029
Fit indexes								
Deviance	993007	932983	925845	910656	1030107	987987	985936	977841
<i>df</i>	424749	424747	424744	424740	424749	424747	424744	424740
AIC	993013	932993	925861	910680	1030113	987997	985952	977865
BIC	993046	933047	925949	910811	1030146	988052	986039	977996

Note: AIC = Akaike information criterion; BIC = Bayesian information criterion.

The coefficients β_{int} and β_{slope} are the estimated average treatment effect of public schools on the intercepts and the slopes, respectively. We estimated these effects using Model 1 as a base, and had implemented different strategies to address selection bias in Models 2 to 7 (for model descriptions, see Table 3). Regarding the confounding variables used to adjust for selection bias, we considered the types of variables described by Bellei (2008). Tables 4 and 5 show that it is relevant to include these variables.

The estimated average treatment effects are presented in Figures 2 and 3; tables with the estimated parameters of Models 1 to 7 can be found in Appendix 3. In Model 1, the differences between voucher and public schools are significant, showing larger intercepts and positive growth for the students from voucher schools. The differences are larger for mathematics than for reading. When control variables are included through linear adjustment (Model 2), matching (Model 3), or both (Model 4), the effect of voucher schools is smaller and positive for the intercepts, and equal or slightly larger and positive for the slopes. If the control is done through propensity score matching, the differences between voucher and public schools are smaller. With Models 2, 3, and 4, we are estimating Type A effects of voucher versus public schools (Raudenbush & Willms, 1995).

Nevertheless, if we want to measure Type B effects, these models are not correct. When we do not include school-level characteristics, the treatment variable is endogenous, because of the system stratification and segregation (Hanchane & Mostafa, 2012). This is reflected in the data, where the most dramatic change in the estimated effects appears when we added the school-level aggregate variables (Models 5, 6, and 7, Figure 2). When these variables are added, the estimated treatment effects on the intercepts become minimal. Besides, for the matching strategies, the estimated effects become non-significant. The same pattern is observed for mathematics and reading. As for the effects on the slopes, when the school characteristics are added, they decrease notably for mathematics but not for reading (Figure 3). To provide a meaningful effect size for the slopes, we calculated the increase in the scores produced by attending a voucher school during 1 year, in comparison with attending a public school. This calculation considers the size of the slope and the scale of the time variable. We found that for each year of education, the mathematics normalized scores of a student in the

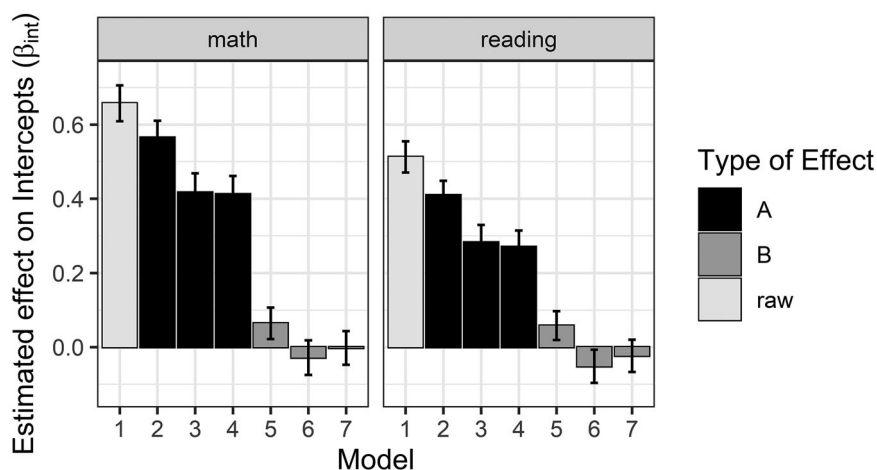


Figure 2. Estimated average treatment effect of voucher education on intercepts.

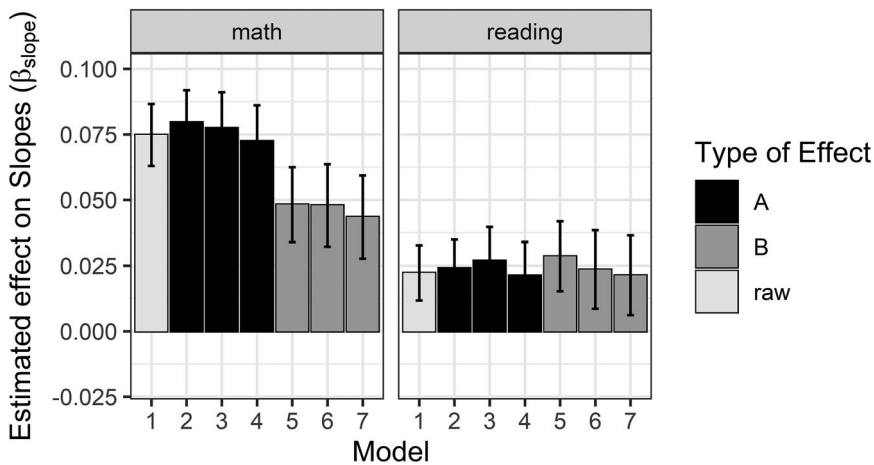


Figure 3. Estimated average treatment effect of voucher education on slopes.

voucher system increase between 0.02 (for the smaller slope) and 0.03 (for the larger slope) times more than the scores of a student in the public system. The reading normalized scores of a student in the voucher system increase about 0.01 times more than the scores of a student in a public school. Considering that the scores are normalized, these increases can be considered as effect sizes.

Discussion and conclusions

Our aim in this article was to measure the effect of voucher and public schools on student trajectories in mathematics and reading. We considered a 6-year period from Grade 4 through to Grade 10, and we compared the trajectories of students who were in the voucher system to students in the public system. Therefore, we measured the effects of being in the voucher school system rather than being in the public school system over a period of 6 years.

These effects are estimated in an educational system that intensely segregates students from public schools and from voucher schools, meaning it is necessary to consider selection bias. We found absolute standardized differences between 0.36 and 0.88 for socioeconomic-related variables and between 0.17 and 0.37 for attendance to pre-school education. Regarding school aggregated variables, the differences are dramatic, with absolute standardized differences ranging between 0.63 and 1.66. Considering common criteria to interpret standardized differences, several of these values can be considered very large, and illustrate the high level of stratification of the Chilean educational system during the years under study.

We modelled student linear trajectories in each discipline. Because the sample size was very large, it has the power to detect very small effects that may not be relevant. To address this issue, for the main conclusions we considered the size and significance of the effects (Funder & Ozer, 2019). We first analysed the effects of public education on the intercepts of the trajectories, which correspond to the test scores in Grade 10, and then we analysed the effect of voucher education on the slopes. Without controlling

for background variables, the estimated effect of voucher schools on students' intercepts is positive and large. The effect is larger for mathematics than for reading. When we controlled for student background variables, the positive effect of voucher schools on students' intercepts became smaller, but was still of substantial size (Models 2, 3, and 4, [Figure 2](#)). These estimations correspond to Type A effects and include the effect of the school context and the school practices (Raudenbush & Willms, 1995). Then, when we controlled for school aggregated variables, the effect of voucher schools changed dramatically and became close to null for mathematics and reading scores (Models 5, 6, and 7, [Figure 2](#)). This pattern occurred for the three estimation methods. These estimates correspond to Type B effects, under the assumption that we do not have endogeneity. It is important to notice that Type A effects are important for families who want effective schools regardless of what explains the differences, and that Type B effects are important for policymakers who want to evaluate school practices and school efficacy.

Concerning student slopes, the effects of voucher schools are positive and significant. They did not change systematically when we controlled for student background variables and school aggregated variables. The slopes show that students in voucher schools get an advantage between 0.02 and 0.03 in mathematics normalized scores and about 0.01 in their reading normalized scores per year. These differences can be considered not relevant in terms of effect sizes for 1 year; however, if we consider several years of education (e.g., the 12 years of compulsory primary and secondary education in Chile), they became relevant.

In summary, this work contributes to an understanding of the effect of the voucher system versus the public system on long-term student trajectories in Chile. Specifically, the study shows that voucher schools have positive Type A effects on student trajectories between Grades 4 and 10 in comparison with public schools; however, if we consider Type B effects (Raudenbush & Willms, 1995) the differences have a very small size for the intercepts. Regarding slopes, the differences are significant in all the considered models, and its relevance in terms of effect sizes depends on the number of years considered. In addition, when matching was used in combination with linear adjusting, the differences in the intercepts are not significant. These results coincide with findings of the OECD (2012) that found no differences between voucher and public schools in the Programme for International Student Assessment results after controlling by socioeconomic and school composition variables. In other words, the differences in trajectories between students in public schools and students in voucher schools were mainly explained by students' characteristics and school context. These results show the considerable influence of school socioeconomic composition and selection practices on student trajectories, because these factors explain most of the observed differences between public and voucher schools. There seem to be small accumulative differences in the effectiveness of practices between the two types of schools favouring voucher schools, but there are quite considerable compositional effects. In short, socioeconomically privileged students are placed in more favourable learning conditions simply because they meet with peers who are themselves privileged. This is evidence of the great inequity of the educational system.

This study analysed the effect of the schools on trajectories, and as such extends the conclusions of studies that have found no relevant or no significant differences between public and voucher schools using one outcome measure (Anand et al., 2009; Bellei, 2008; Lara et al., 2011; Zubizarreta & Keele, 2017).

The general conclusions are the same, regardless of the methods used (propensity score matching, linear adjustment, or both methods combined). However, there are some differences in the size of the effects. The estimated positive effect of the voucher schools was larger with linear adjusting in the models that only controlled for student background variables. This fact can be explained by two reasons. First, it could be an effect of extrapolating results for a large number of students who are not comparable. Second, it could be an effect of including only linear effects of the control variables.

Regarding propensity score matching, the estimated effects confirm the results found with the other methods. Moreover, they show an enormous group of students who are not comparable. [Figure 1](#) shows summarily that public schools are in charge of educating the students with the most disadvantaged socioeconomic conditions. Thus, it is not possible to measure the effect of voucher schools in the group of students with the most disadvantaged socioeconomic conditions, because these schools are not educating this population. Similarly, we cannot measure the efficacy of public schools in the group of students with more advantaged socioeconomic conditions.

Bettinger (2011) states that, in order to draw up policy recommendations regarding the effectiveness of a voucher system, it is important to understand the mechanisms underlying it. In this study, it appears that two of the mechanisms that explain voucher schools' better achievement are peer effects and the ability of schools to select students. This finding is not in line with the usual argument to support voucher schools in Chile, which suggests that school competition will produce more efficacious schools.

A limitation of this study is that it considers that the assignment process to voucher and public schools depends only on observed variables. Other methods, which do not have these hypotheses, could be used, such as instrumental variables.

Regarding the practical implications, special attention should be given to public schools because they are educating students with more disadvantaged background conditions. The segregation of the system is harming these students because when compositional effects are not controlled, they have lower test scores than voucher school students. If we want to stop impairing the most disadvantaged students, the mechanisms producing this segregation, such as school selection practices, should be eliminated, or at least strongly reduced.

Finally, it should be noted that Chile is changing its educational policies in order to move towards an integrated system of public education with several characteristics which aim to reduce segregation (Bellei, 2018). Once these policies are properly enacted, it would be interesting to replicate this study in order to understand the effect of the new policies on the system.

Acknowledgements

The authors thank Agencia de Calidad de la Educación, Gobierno de Chile for the access to the data. All the results of the study are the authors' responsibility.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This work was partially supported by ANID/PIA/Basal Funds for Centers of Excellence FB0003, Apoyo a Centros de Excelencia ACE210010, Proyecto Fondo Basal FB210005, and Beca Doctorado Nacional 2013 Grant number 21130684. This research/thesis was partially supported by the supercomputing infrastructure of the NLHPC (ECM-02).

Data availability statement

The data that support the findings of this study are available from Agencia de Calidad de la Educación, Gobierno de Chile. Restrictions apply to the availability of these data, which were used under license for this study.

Notes on contributors

Valentina Giacconi is an assistant professor at the University of O'Higgins in Chile. She has a PhD in Educational Sciences from the University Grenoble Alpes, France, and mathematical modelling from the University of Chile, Chile. She specialized on the application of statistical methods for research in education.

Pascal Bressoux is a full professor at Université Grenoble Alpes. His research interests are in educational evaluation and the impact of teaching methods on students' achievement. He specialized in multilevel modelling and longitudinal data analysis.

Patricio Felmer has a PhD in Mathematics from the University of Wisconsin-Madison in the USA. He dedicated the first part of his academic life to mathematical research. In a second part, he has directed his interests to education, and, together with collaborators, he founded the ARPA Initiative, a teacher professional development programme, whose purpose is to incorporate problem solving in Chilean classrooms.

ORCID

Valentina Giacconi  <http://orcid.org/0000-0002-5166-5673>

Pascal Bressoux  <http://orcid.org/0000-0001-8018-5612>

References

- Agencia de Calidad de la Educación. (n.d.). *Metodología de construcción de grupos socioeconómicos. Pruebas SIMCE 2012* [Socioeconomic group construction methodology. SIMCE tests 2012] (Technical Report.). Gobierno de Chile.
- Agencia de Calidad de la Educación. (2015). *Informe técnico SIMCE 2013* [SIMCE 2013 Technical report]. Gobierno de Chile.
- Anand, P., Mizala, A., & Repetto, A. (2009). Using school scholarships to estimate the effect of private education on the academic achievement of low-income students in Chile. *Economics of Education Review*, 28(3), 370–381. <https://doi.org/10.1016/j.econedurev.2008.03.005>
- Angrist, J., Bettinger, E., & Kremer, M. (2006). Long-term educational consequences of secondary school vouchers: Evidence from administrative records in Colombia. *American Economic Review*, 96(3), 847–862. <https://doi.org/10.1257/aer.96.3.847>
- Austin, P. C. (2011). Optimal caliper widths for propensity-score matching when estimating differences in means and differences in proportions in observational studies. *Pharmaceutical Statistics*, 10(2), 150–161. <https://doi.org/10.1002/pst.433>
- Bates, D., Mächler, M., Bolker, B., & Walker S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1), 1–48. <https://doi.org/10.18637/jss.v067.i01>

- Bellei, C. (2008). The public-private school controversy in Chile. In R. Chakrabarti & P. E. Peterson (Eds.), *School choice international: Exploring public-private partnerships* (pp. 165–192). MIT Press.
- Bellei, C. (2018). La más grande reforma institucional en educación desde el retorno a la democracia [The largest institutional reform in education since the return to democracy]. In C. Bellei (Ed.), *Nueva Educación Pública: Contexto, contenidos y perspectivas de la desmunicipalización* (pp. 11–16). Universidad de Chile.
- Bettinger, E. (2011). Educational vouchers in international contexts. In E. A. Hanushek, S. Machin, & L. Woessmann (Eds.), *Handbook of the economics of education* (Vol. 4, pp. 551–572). North-Holland.
- Bettinger, E., Kremer, M., Kugler, M., Medina-Durango, C. A., Posso-Suárez, C. M., Saavedra, J. E., & Medina, C. (2019). *School vouchers, labor markets and vocational education* (Borradores de Economía No. 1087). Banco de la República de Colombia.
- Betts, J. R., & Tang, Y. E. (2019). The effects of charter schools on student achievement. In M. Berends, R. J. Waddington, & J. Schoenig (Eds.), *School choice at the crossroads: Research perspectives* (pp. 69–91). Routledge.
- Böhlmark, A., & Lindahl, M. (2015). Independent schools and long-run educational outcomes: Evidence from Sweden's large-scale voucher reform. *Economica*, 82(327), 508–551. <https://doi.org/10.1111/ecca.12130>
- Bressoux, P. (2010). *Modélisation statistique appliquée aux sciences sociales* [Statistical modeling applied to social sciences] (2nd ed.). De Boeck Supérieur. <https://doi.org/10.3917/dbu.bress.2010.01>
- Correa, J. A., Parro, F., & Reyes, L. (2014). The effects of vouchers on school results: Evidence from Chile's targeted voucher program. *Journal of Human Capital*, 8(4), 351–398. <https://doi.org/10.1086/679282>
- Crespo-Cebada, E., Pedraja-Chaparro, F., & Santín, D. (2014). Does school ownership matter? An unbiased efficiency comparison for regions of Spain. *Journal of Productivity Analysis*, 41(1), 153–172. <https://doi.org/10.1007/s11123-013-0338-y>
- Enders, C. K., Mistler, S. A., & Keller, B. T. (2016). Multilevel multiple imputation: A review and evaluation of joint modeling and chained equations imputation. *Psychological Methods*, 21(2), 222–240. <https://doi.org/10.1037/met0000063>
- Epple, D., Romano, R. E., & Urquiola, M. (2017). School vouchers: A survey of the economics literature. *Journal of Economic Literature*, 55(2), 441–492. <https://doi.org/10.1257/jel.20150679>
- Funder, D. C., & Ozer, D. J. (2019). Evaluating effect size in psychological research: Sense and non-sense. *Advances in Methods and Practices in Psychological Science*, 2(2), 156–168. <https://doi.org/10.1177/2515245919847202>
- Guldemond, H., & Bosker, R. J. (2009). School effects on students' progress – A dynamic perspective. *School Effectiveness and School Improvement*, 20(2), 255–268. <https://doi.org/10.1080/09243450902883938>
- Hanchane, S., & Mostafa, T. (2012). Solving endogeneity problems in multilevel estimation: An example using education production functions. *Journal of Applied Statistics*, 39(5), 1101–1114. <https://doi.org/10.1080/02664763.2011.638705>
- Hofflinger, A., & von Hippel, P. T. (2020). Does achievement rise fastest with school choice, school resources, or family resources? Chile from 2002 to 2013. *Sociology of Education*, 93(2), 132–152. <https://doi.org/10.1177/0038040719899358>
- Lara, B., Mizala, A., & Repetto, A. (2011). The effectiveness of private voucher education: Evidence from structural school switches. *Educational Evaluation and Policy Analysis*, 33(2), 119–137. <https://doi.org/10.3102/0162373711402990>
- Law N° 20845. (2015). Diario Oficial de la República de Chile, Santiago, Chile, June 8, 2015. <https://www.leychile.cl/Navegar?idNorma=1078172>
- Manzi, J., San Martín, E., & Van Belleghem, S. (2014). School system evaluation by value added analysis under endogeneity. *Psychometrika*, 79(1), 130–153. <https://doi.org/10.1007/s11336-013-9338-0>
- Ministerio de Educación. (2014). *Estadísticas de la educación 2013* [Educational Statistics 2013]. <https://bibliotecadigital.mineduc.cl/handle/20.500.12365/2407>

- Mizala, A., & Torche, F. (2017). Means-tested school vouchers and educational achievement: Evidence from Chile's universal voucher system. *The ANNALS of the American Academy of Political and Social Science*, 674(1), 163–183. <https://doi.org/10.1177/0002716217732033>
- Morgan, C., Petrosino, A., & Fronius, T. (2015). The impact of school vouchers in developing countries: A systematic review. *International Journal of Educational Research*, 72, 70–79. <https://doi.org/10.1016/j.ijer.2015.04.010>
- Organisation for Economic Co-operation and Development. (2012). *Public and private schools: How management and funding relate to their socio-economic profile*. <https://doi.org/10.1787/9789264175006-en>
- Ortega Ferrand, L. C. (2016). *Educational effectiveness and inequalities in Chile: A multilevel accelerated longitudinal study of primary school children's achievement trajectories* [Unpublished doctoral dissertation]. University of Oxford.
- Pattanayak, C. W. (2015). Evaluating covariate balance. In W. Pan & H. Bai (Eds.), *Propensity score analysis: Fundamentals and developments* (pp. 89–112). The Guilford Press.
- Raczynski, D., Muñoz, G., Weinstein, J., & Pascual, J. (2013). Subvención escolar preferencial (SEP) en Chile: Un intento para equilibrar la macro y la micro política escolar [Preferential school subsidy in Chile: Effort to balance macro and micro educational policy]. *REICE. Revista Electrónica Iberoamericana sobre Calidad, Eficacia y Cambio en Educación*, 11(2), 164–193. <https://revistas.uam.es/reice/article/view/2902>
- Raudenbush, S. W., & Willms, J. D. (1995). The estimation of school effects. *Journal of Educational and Behavioral Statistics*, 20(4), 307–335. <https://doi.org/10.2307/1165304>
- Rubin, D. B. (1976). Inference and missing data. *Biometrika*, 63(3), 581–592. <https://doi.org/10.2307/2335739>
- Sekhon, J. S. (2011). Multivariate and propensity score matching software with automated balance optimization: The Matching package for R. *Journal of Statistical Software*, 42(7), 1–52. <https://doi.org/10.18637/jss.v042.i07>
- Singer, J. D., & Willett, J. B. (2003). *Applied longitudinal data analysis: Modeling change and event occurrence*. Oxford University Press.
- Valenzuela, J. P., Bellei, C., & de los Ríos, D. (2014). Socioeconomic school segregation in a market-oriented educational system. The case of Chile. *Journal of Education Policy*, 29(2), 217–241. <https://doi.org/10.1080/02680939.2013.806995>
- Valenzuela, J. P., Villarroel, G., & Villalobos, C. (2013). Ley de Subvención Escolar Preferencial (SEP): Algunos resultados preliminares de su implementación [Preferential School Subsidy Law (SEP): Some preliminary results of its implementation]. *Pensamiento Educativo: Revista de Investigación Educativa Latinoamericana*, 50(2), 113–131. <http://pensamientoeducativo.uc.cl/index.php/pel/article/view/26233>
- van Buuren, S. (2011). Multiple imputation of multilevel data. In J. J. Hox & J. K. Roberts (Eds.), *Handbook of advanced multilevel analysis* (pp. 173–196). Routledge.
- Weinstein, J., & Muñoz, G. (2014). When duties are not enough: Principal leadership and public or private school management in Chile. *School Effectiveness and School Improvement*, 25(4), 651–670. <https://doi.org/10.1080/09243453.2013.792850>
- Zhao, J. H., & Schafer, J. L. (2016). *pan: Multiple imputation for multivariate panel or clustered data* [Computer software manual]. (R package Version 1.4).
- Zimmer, R., Buddin, R., Smith, S. A., & Duffy, D. (2019). *Nearly three decades into the charter school movement, what has research told us about charter schools* (EdWorkingPaper No. 19-156). Annenberg Institute at Brown University. <https://doi.org/10.26300/wkij-3559>
- Zubizarreta, J. R., & Keele, L. (2017). Optimal multilevel matching in clustered observational studies: A case study of the effectiveness of private schools under a large-scale voucher system. *Journal of the American Statistical Association*, 112(518), 547–560. <https://doi.org/10.1080/01621459.2016.1240683>

Appendix 1. Descriptive statistics of students in public, voucher, and private schools

Variable	2007 – Grade 4 Mean (SD)			2011 – Grade 8 Mean (SD)			2013 – Grade 10 Mean (SD)		
	Public	Voucher	Private	Public	Voucher	Private	Public	Voucher	Private
<i>N</i>	68,433	75,868	13,513	64,749	78,920	14,145	55,762	87,272	14,780
Years of father education	10.06 (3.36)	12.41 (3.32)	16.72 (2.03)	9.90 (3.37)	12.22 (3.40)	16.79 (2.07)	9.94 (3.44)	11.96 (3.52)	16.73 (2.11)
Years of mother education	10.08 (3.29)	12.39 (3.19)	16.25 (1.79)	9.98 (3.33)	12.25 (3.25)	16.29 (1.84)	10.07 (3.37)	12.02 (3.35)	16.25 (1.93)
Average income ^a	2.53 (1.58)	4.00 (2.52)	10.98 (2.64)	2.83 (1.72)	4.45 (2.79)	12.54 (3.18)	3.32 (1.91)	4.89 (2.92)	12.77 (3.14)
% with access to a computer	0.36 (0.48)	0.63 (0.48)	0.97 (0.16)	0.77 (0.42)	0.89 (0.31)	1.00 (0.06)	0.82 (0.38)	0.91 (0.29)	1.00 (0.06)
% with access to the internet	0.11 (0.32)	0.32 (0.47)	0.89 (0.31)	0.49 (0.50)	0.71 (0.45)	0.98 (0.13)	0.55 (0.50)	0.75 (0.43)	0.99 (0.12)
Average number of books	21.61 (26.74)	33.97 (33.49)	73.24 (41.54)	31.12 (29.40)	43.89 (35.23)	82.61 (38.90)	34.13 (31.03)	45.76 (35.78)	83.44 (38.92)
Fraction that attended to ECE (0–2 years)	0.04 (0.20)	0.08 (0.27)	0.14 (0.35)	0.12 (0.33)	0.17 (0.38)	0.23 (0.42)	0.11 (0.31)	0.13 (0.34)	0.19 (0.39)
Fraction that attended to ECE (2–4 years)	0.25 (0.44)	0.35 (0.48)	0.71 (0.45)	0.35 (0.48)	0.41 (0.49)	0.69 (0.46)	0.47 (0.50)	0.53 (0.50)	0.85 (0.36)
Fraction that attended to Pre-kindergarten	0.40 (0.49)	0.48 (0.50)	0.54 (0.50)	0.51 (0.50)	0.59 (0.49)	0.88 (0.32)	0.58 (0.49)	0.68 (0.47)	0.93 (0.25)
Fraction that attended to Kindergarten	0.32 (0.46)	0.32 (0.47)	0.36 (0.48)	0.66 (0.47)	0.76 (0.42)	0.96 (0.20)	0.92 (0.27)	0.97 (0.17)	1.00 (0.06)

Note: For each year and all the variables, the differences between public, voucher, and private schools are significant with $p < 0.01$. ECE = early childhood education.

^aThousands of Chilean pesos.

Appendix 2. Description of variables and percentage of missing data

Variable	Level	% Missing
Time	occasion	0
Mother education		23
Father education		26
Number of books		22
Home Income		23
Computer		23
Internet		27
Parents' expectations		25
Normalized score reading		7
Normalized score mathematics		7
SchoolSystem	student	0
Gender		0
Kindergarten		48
Pre-kindergarten		40
ECE (0–2 years)	school (Grade 4)	33
ECE (2–4 years)		27
SES index (Grade 4)		16
SES selection index (Grade 4)		16
Academic selection index (Grade 4)		16
Preschool attendance index (Grade 4)		16
SES index (Grade 8)		1
SES selection index (Grade 8)		1
Academic selection index (Grade 8)		1
Preschool attendance index (Grade 8)		1
SES index (Grade 10)	school (Grade 10)	3
SES selection index (Grade 10)		3
Academic selection index (Grade 10)		3
Preschool attendance index (Grade 10)		3

Note: The percentage of missing data is measured according to the given level; for example, if it is at the student level, the percentage corresponds to students with missing values. ECE = early childhood education; SES = socioeconomic status.

Appendix 3. Estimated parameters for multilevel growth models

In Tables A1–A4, for each of the imputed data bases, the parameters were estimated and then pooled according to Rubin's rules (Rubin, 1976). For random effects and fit indexes, we present also the 5th and 95th percentile to represent variability across imputed data.

Table A1. Fixed effects, random effects, and fit indexes of Models 1, 2, 3, and 4 in mathematics scores.

Parameters	Normalized mathematics score			
	Model 1 β (SE)	Model 2 β (SE)	Model 3 β (SE)	Model 4 β (SE)
Fixed Effects				
Intercept	0.164 (0.015)	−0.56 (0.028)	0.059 (0.016)	−0.595 (0.036)
Time	0.024 (0.004)	−0.05 (0.013)	0.021 (0.004)	−0.072 (0.017)
SchoolSystem	0.657 (0.025)	0.565 (0.023)	0.417 (0.026)	0.412 (0.025)
Time*SchoolSystem	0.075 (0.006)	0.08 (0.006)	0.077 (0.007)	0.072 (0.007)
ECE (0–2 years)		0.017 (0.011)		0.018 (0.017)
Time*ECE (0–2 years)		−0.001 (0.005)		−0.005 (0.008)
ECE (2–4 years)		−0.019 (0.008)		−0.019 (0.011)
Time*ECE (2–4 years)		0.002 (0.004)		0.004 (0.005)
Pre-kinder		−0.031 (0.009)		−0.035 (0.011)
Time*Pre-kinder		−0.003 (0.004)		−0.003 (0.005)
Kinder		0.058 (0.009)		0.048 (0.013)
Time*Kinder		0.014 (0.004)		0.014 (0.005)
Female		−0.22 (0.005)		−0.231 (0.008)
Time*Female		−0.021 (0.002)		−0.02 (0.004)

(Continued)

Table A1. Continued.

Parameters	Normalized mathematics score			
	Model 1 β (SE)	Model 2 β (SE)	Model 3 β (SE)	Model 4 β (SE)
Mother education		0.005 (0.001)		0.004 (0.001)
Time*Mother education		−0.002 (0)		−0.002 (0.001)
Father education		0.006 (0.001)		0.005 (0.001)
Time*Father education		0 (0)		−0.001 (0.001)
Parents' expectations		0.037 (0.001)		0.037 (0.002)
Time*Parents' expectations		0.004 (0.001)		0.006 (0.001)
Number of books		0.001 (0)		0.001 (0)
Time*Number of books		0 (0)		0 (0)
Home income		0.004 (0.001)		0.002 (0.001)
Time*Home income		0.002 (0.001)		0.001 (0.001)
Computer		0.046 (0.007)		0.061 (0.009)
Time*Computer		0.008 (0.004)		0.017 (0.005)
Internet		−0.03 (0.005)		−0.041 (0.007)
Time*Internet		0.006 (0.004)		−0.003 (0.006)
Random Intercepts				
Residual mean	0.245	0.247	0.245	0.247
Residual P5	0.244	0.246	0.244	0.246
Residual P95	0.246	0.247	0.247	0.248
Student mean	0.377	0.345	0.369	0.339
Student P5	0.374	0.344	0.366	0.336
Student P95	0.378	0.346	0.371	0.341
School (Grade 8) mean	0.04	0.038	0.038	0.037
School (Grade 8) P5	0.04	0.037	0.038	0.036
School (Grade 8) P95	0.041	0.038	0.039	0.038
School (Grade 10) mean	0.279	0.242	0.266	0.239
School (Grade 10) P5	0.277	0.24	0.264	0.236
School (Grade 10) P95	0.281	0.243	0.269	0.241
Random Slopes				
Student-time mean	0.021	0.02	0.019	0.018
Student-time P5	0.02	0.02	0.018	0.018
Student-time P95	0.021	0.02	0.019	0.019
School (Grade 8)-time mean	0.008	0.008	0.008	0.008
School (Grade 8)-time P5	0.008	0.008	0.008	0.008
School (Grade 8)-time P95	0.008	0.008	0.008	0.008
School (Grade 10)-time mean	0.011	0.01	0.01	0.01
School (Grade 10)-time P5	0.01	0.01	0.01	0.01
School (Grade 10)-time P95	0.011	0.01	0.011	0.01
Fit Indexes				
Deviance mean	624,520	619,190	289,260	286,896
Deviance P5	624,234	618,858	289,060	286,694
Deviance P95	624,755	619,465	289,544	287,108
df	290,908	290,884	134,038	134,014
AIC mean	624,548	619,266	289,288	286,973
AIC P5	624,262	618,935	289,088	286,770
AIC P95	624,783	619,541	289,572	287,184
BIC mean	624,697	619,668	289,425	287,345
BIC P5	624,410	619,337	289,226	287,142
BIC P95	624,931	619,944	289,709	287,556

Note: ECE = early childhood education; P5 = 5th percentile among imputations; P95 = 95th percentile among imputations; AIC = Akaike information criterion; BIC = Bayesian information criterion.

Table A2. Fixed effects, random effects, and fit indexes of Models 5, 6, and 7 in mathematics scores.

	Normalized mathematics score		
	Model 5 β (SE)	Model 6 β (SE)	Model 7 β (SE)
Fixed Effects			
Intercept	−0.62 (0.027)	−0.01 (0.014)	−0.612 (0.035)
Time	−0.058 (0.014)	0.024 (0.005)	−0.079 (0.017)

(Continued)

Table A2. Continued.

	Normalized mathematics score		
	Model 5 β (SE)	Model 6 β (SE)	Model 7 β (SE)
SchoolSystem	0.064 (0.022)	−0.028 (0.024)	−0.002 (0.023)
Time*SchoolSystem	0.048 (0.007)	0.048 (0.008)	0.043 (0.008)
ECE (0–2 years)	0.017 (0.011)		0.017 (0.017)
Time*ECE (0–2 years)	−0.001 (0.005)		−0.005 (0.007)
ECE (2–4 years)	−0.018 (0.008)		−0.017 (0.011)
Time*ECE (2–4 years)	0.003 (0.004)		0.004 (0.005)
Pre-kinder	−0.031 (0.009)		−0.034 (0.011)
Time*Pre-kinder	−0.003 (0.004)		−0.003 (0.005)
Kinder	0.056 (0.009)		0.045 (0.013)
Time*Kinder	0.014 (0.004)		0.014 (0.005)
Female	−0.221 (0.005)		−0.232 (0.008)
Time*Female	−0.021 (0.002)		−0.019 (0.004)
Mother education	0.003 (0.001)		0.003 (0.001)
Time*Mother education	−0.002 (0)		−0.001 (0.001)
Father education	0.005 (0.001)		0.003 (0.001)
Time*Father education	0.001 (0)		0 (0.001)
Parents' expectations	0.036 (0.001)		0.036 (0.002)
Time*Parents' expectations	0.005 (0.001)		0.006 (0.001)
Number of books	0.001 (0)		0.001 (0)
Time*Number of books	0 (0)		0 (0)
Home income	0.003 (0.001)		0.001 (0.001)
Time*Home income	0.002 (0.001)		0.002 (0.001)
Computer	0.045 (0.007)		0.059 (0.009)
Time*Computer	0.009 (0.004)		0.018 (0.005)
Internet	−0.032 (0.005)		−0.044 (0.007)
Time*Internet	0.007 (0.004)		−0.002 (0.006)
SES index (Grade 8)	0.084 (0.011)	0.092 (0.014)	0.066 (0.014)
Time*SES index (Grade 8)	−0.027 (0.005)	−0.035 (0.006)	−0.032 (0.006)
SES index (Grade 10)	0.424 (0.017)	0.438 (0.021)	0.409 (0.02)
Time*SES index (Grade 10)	0.042 (0.006)	0.039 (0.007)	0.036 (0.007)
Preschool attendance index (Grade 8)	−0.057 (0.007)	−0.065 (0.01)	−0.062 (0.01)
Time*Preschool attendance index (Grade 8)	0 (0.003)	0.001 (0.004)	0.001 (0.004)
Preschool attendance index (Grade 10)	−0.159 (0.013)	−0.171 (0.016)	−0.161 (0.015)
Time*Preschool attendance index (Grade 10)	−0.014 (0.004)	−0.015 (0.005)	−0.014 (0.005)
Academic selection index (Grade 8)	0.02 (0.008)	0.041 (0.01)	0.04 (0.01)
Time*Academic selection index (Grade 8)	0.001 (0.003)	0.006 (0.004)	0.006 (0.004)
Academic selection index (Grade 10)	0.118 (0.011)	0.119 (0.012)	0.116 (0.012)
Time*Academic selection index (Grade 10)	0.018 (0.003)	0.016 (0.004)	0.016 (0.004)
SES selection index (Grade 8)	0.013 (0.006)	0.011 (0.009)	0.017 (0.009)
Time*SES selection index (Grade 8)	−0.006 (0.003)	−0.006 (0.004)	−0.005 (0.004)
SES selection index (Grade 10)	0.11 (0.012)	0.122 (0.015)	0.118 (0.015)
Time*SES selection index (Grade 10)	0.022 (0.004)	0.024 (0.005)	0.024 (0.005)
Random Intercepts			
Residual mean	0.246	0.245	0.247
Residual P5	0.246	0.244	0.246
Residual P95	0.247	0.247	0.248
Student mean	0.346	0.369	0.339
Student P5	0.344	0.366	0.337
Student P95	0.347	0.37	0.341
School (Grade 8) mean	0.036	0.036	0.035
School (Grade 8) P5	0.035	0.035	0.034
School (Grade 8) P95	0.036	0.036	0.036
School (Grade 10) mean	0.122	0.133	0.124
School (Grade 10) P5	0.12	0.131	0.123
School (Grade 10) P95	0.123	0.135	0.125
Random Slopes			
Student-time mean	0.02	0.019	0.018
Student-time P5	0.02	0.018	0.018
Student-time P95	0.02	0.019	0.019

(Continued)

Table A2. Continued.

	Normalized mathematics score		
	Model 5 β (SE)	Model 6 β (SE)	Model 7 β (SE)
School (Grade 8)-time mean	0.008	0.008	0.008
School (Grade 8)-time P5	0.008	0.008	0.008
School (Grade 8)-time P95	0.008	0.008	0.008
School (Grade 10)-time mean	0.009	0.009	0.009
School (Grade 10)-time P5	0.009	0.009	0.009
School (Grade 10)-time P95	0.009	0.01	0.009
Fit Indexes			
Deviance mean	617,345	287,753	285,531
Deviance P5	617,042	287,556	285,327
Deviance P95	617,619	288,023	285,738
df	290,868	134,022	133,998
AIC mean	617,453	287,813	285,639
AIC P5	617,150	287,616	285,435
AIC P95	617,727	288,083	285,846
BIC mean	618,024	288,107	286,168
BIC P5	617,722	287,910	285,965
BIC P95	618,299	288,377	286,375

Note: ECE = early childhood education; SES = socioeconomic status; P5 = 5th percentile among imputations; P95 = 95th percentile among imputations; AIC = Akaike information criterion; BIC = Bayesian information criterion.

Table A3. Fixed effects, random effects, and fit indexes of Models 1, 2, 3, and 4 in reading scores.

	Normalized reading score			
	Model 1 β (SE)	Model 2 β (SE)	Model 3 β (SE)	Model 4 β (SE)
Fixed Effects				
Intercept	0.153 (0.013)	−0.859 (0.027)	0.054 (0.014)	−0.858 (0.038)
Time	0.013 (0.003)	−0.126 (0.013)	0.016 (0.004)	−0.124 (0.017)
SchoolSystem	0.513 (0.021)	0.41 (0.02)	0.282 (0.024)	0.27 (0.023)
Time*SchoolSystem	0.022 (0.005)	0.024 (0.006)	0.027 (0.007)	0.021 (0.007)
ECE (0–2 years)		0.008 (0.013)		−0.005 (0.02)
Time*ECE (0–2 years)		−0.005 (0.005)		−0.013 (0.009)
ECE (2–4 years)		−0.021 (0.009)		−0.015 (0.013)
Time*ECE (2–4 years)		0.004 (0.004)		0.008 (0.005)
Pre-kinder		−0.037 (0.01)		−0.042 (0.012)
Time*Pre-kinder		−0.004 (0.004)		−0.004 (0.006)
Kinder		0.063 (0.009)		0.058 (0.012)
Time*Kinder		0.014 (0.005)		0.014 (0.007)
Female		0.163 (0.006)		0.164 (0.009)
Time*Female		0.032 (0.003)		0.036 (0.004)
Mother education		0.005 (0.001)		0.004 (0.001)
Time*Mother education		−0.002 (0.001)		−0.001 (0.001)
Father education		0.006 (0.001)		0.005 (0.001)
Time*Father education		0.001 (0.001)		−0.001 (0.001)
Parents' expectations		0.04 (0.001)		0.039 (0.002)
Time*Parents' expectations		0.006 (0.001)		0.006 (0.001)
Number of books		0.001 (0)		0.001 (0)
Time*Number of books		0 (0)		0 (0)
Home income		0 (0.001)		−0.002 (0.002)
Time*Home income		0 (0.001)		−0.001 (0.001)
Computer		0.07 (0.008)		0.076 (0.01)
Time*Computer		0.028 (0.005)		0.035 (0.006)
Internet		−0.035 (0.007)		−0.037 (0.008)
Time*Internet		−0.01 (0.004)		−0.013 (0.007)
Random Intercepts				
Residual mean	0.315	0.317	0.313	0.315
Residual P5	0.314	0.316	0.312	0.313
Residual P95	0.316	0.318	0.314	0.316
Student mean	0.424	0.394	0.411	0.383

(Continued)

Table A3. Continued.

	Normalized reading score			
	Model 1 β (SE)	Model 2 β (SE)	Model 3 β (SE)	Model 4 β (SE)
Student P5	0.422	0.392	0.408	0.38
Student P95	0.426	0.396	0.413	0.386
School (Grade 8) mean	0.029	0.026	0.031	0.029
School (Grade 8) P5	0.028	0.026	0.03	0.028
School (Grade 8) P95	0.03	0.027	0.033	0.03
School (Grade 10) mean	0.203	0.164	0.202	0.172
School (Grade 10) P5	0.2	0.162	0.199	0.169
School (Grade 10) P95	0.205	0.166	0.204	0.175
Random Slopes				
Student-time mean	0.013	0.013	0.013	0.012
Student-time P5	0.013	0.012	0.012	0.011
Student-time P95	0.013	0.013	0.013	0.013
School (Grade 8)-time mean	0.005	0.005	0.006	0.006
School (Grade 8)-time P5	0.005	0.005	0.005	0.005
School (Grade 8)-time P95	0.005	0.005	0.006	0.006
School (Grade 10)-time mean	0.008	0.008	0.008	0.008
School (Grade 10)-time P5	0.008	0.007	0.008	0.008
School (Grade 10)-time P95	0.008	0.008	0.008	0.008
Fit Indexes				
Deviance mean	670,932	666,619	309,571	307,732
Deviance P5	670,687	666,343	309,409	307,502
Deviance P95	671,282	666,985	309,765	307,918
df	290,908	290,884	134,038	134,014
AIC mean	670,960	666,695	309,599	307,808
AIC P5	670,715	666,419	309,437	307,578
AIC P95	671,310	667,061	309,793	307,994
BIC mean	671,108	667,098	309,736	308,180
BIC P5	670,864	666,821	309,574	307,950
BIC P95	671,458	667,463	309,930	308,367

Note: ECE = early childhood education; P5 = 5th percentile among imputations; P95 = 95th percentile among imputations; AIC = Akaike information criterion; BIC = Bayesian information criterion.

Table A4. Fixed effects, random effects, and fit indexes of Models 5, 6, and 7 in reading scores.

	Normalized reading score		
	Model 5 β (SE)	Model 6 β (SE)	Model 7 β (SE)
Fixed Effects			
Intercept	−0.888 (0.026)	0.013 (0.014)	−0.855 (0.037)
Time	−0.135 (0.013)	0.02 (0.005)	−0.137 (0.018)
SchoolSystem	0.058 (0.02)	−0.051 (0.023)	−0.023 (0.022)
Time*SchoolSystem	0.029 (0.007)	0.024 (0.008)	0.021 (0.008)
ECE (0–2 years)	0.008 (0.013)		−0.005 (0.02)
Time*ECE (0–2 years)	−0.005 (0.005)		−0.013 (0.009)
ECE (2–4 years)	−0.02 (0.009)		−0.011 (0.013)
Time*ECE (2–4 years)	0.004 (0.004)		0.008 (0.005)
Pre-kinder	−0.036 (0.01)		−0.04 (0.012)
Time*Pre-kinder	−0.004 (0.004)		−0.004 (0.006)
Kinder	0.062 (0.009)		0.056 (0.012)
Time*Kinder	0.015 (0.005)		0.015 (0.007)
Female	0.162 (0.006)		0.162 (0.009)
Time*Female	0.032 (0.003)		0.037 (0.004)
Mother education	0.004 (0.001)		0.003 (0.001)
Time*Mother education	−0.001 (0.001)		−0.001 (0.001)
Father education	0.006 (0.001)		0.004 (0.001)
Time*Father education	0.001 (0.001)		0 (0.001)
Parents' expectations	0.04 (0.001)		0.039 (0.002)
Time*Parents' expectations	0.006 (0.001)		0.006 (0.001)
Number of books	0.001 (0)		0.001 (0)

(Continued)

Table A4. Continued.

	Normalized reading score		
	Model 5 β (SE)	Model 6 β (SE)	Model 7 β (SE)
Time*Number of books	0 (0)		0 (0)
Home income	0 (0.001)		−0.003 (0.002)
Time*Home income	0 (0.001)		0 (0.001)
Computer	0.069 (0.008)		0.075 (0.01)
Time*Computer	0.03 (0.005)		0.036 (0.006)
Internet	−0.036 (0.007)		−0.038 (0.008)
Time*Internet	−0.009 (0.004)		−0.011 (0.007)
SES index (Grade 8)	0.037 (0.011)	0.054 (0.015)	0.035 (0.015)
Time*SES index (Grade 8)	−0.027 (0.005)	−0.023 (0.006)	−0.02 (0.006)
SES index (Grade 10)	0.353 (0.016)	0.386 (0.02)	0.342 (0.02)
Time*SES index (Grade 10)	0.01 (0.006)	0.009 (0.007)	0.002 (0.007)
Preschool attendance index (Grade 8)	−0.05 (0.007)	−0.059 (0.01)	−0.051 (0.01)
Time*Preschool attendance index (Grade 8)	0.005 (0.003)	0.005 (0.004)	0.006 (0.004)
Preschool attendance index (Grade 10)	−0.158 (0.013)	−0.183 (0.016)	−0.17 (0.015)
Time*Preschool attendance index (Grade 10)	−0.006 (0.004)	−0.008 (0.006)	−0.007 (0.006)
Academic selection index (Grade 8)	0.009 (0.008)	0.023 (0.01)	0.023 (0.01)
Time*Academic selection index (Grade 8)	0.002 (0.003)	0.003 (0.004)	0.003 (0.004)
Academic selection index (Grade 10)	0.081 (0.01)	0.082 (0.012)	0.08 (0.012)
Time*Academic selection index (Grade 10)	0.003 (0.003)	0.001 (0.004)	0.001 (0.004)
SES selection index (Grade 8)	0.006 (0.006)	0.012 (0.009)	0.008 (0.009)
Time*SES selection index (Grade 8)	−0.008 (0.003)	−0.009 (0.004)	−0.009 (0.004)
SES selection index (Grade 10)	0.09 (0.011)	0.103 (0.015)	0.099 (0.015)
Time*SES selection index (Grade 10)	0.012 (0.004)	0.015 (0.005)	0.014 (0.005)
Random Intercepts			
Residual mean	0.317	0.314	0.314
Residual P5	0.316	0.312	0.313
Residual P95	0.318	0.315	0.316
Student mean	0.395	0.41	0.384
Student P5	0.392	0.406	0.381
Student P95	0.397	0.413	0.386
School (Grade 8) mean	0.025	0.03	0.028
School (Grade 8) P5	0.024	0.029	0.027
School (Grade 8) P95	0.026	0.031	0.03
School (Grade 10) mean	0.096	0.115	0.104
School (Grade 10) P5	0.094	0.113	0.102
School (Grade 10) P95	0.098	0.117	0.106
Random Slopes			
Student-time mean	0.013	0.012	0.012
Student-time P5	0.012	0.011	0.011
Student-time P95	0.013	0.013	0.012
School (Grade 8)-time mean	0.005	0.005	0.005
School (Grade 8)-time P5	0.005	0.005	0.005
School (Grade 8)-time P95	0.005	0.006	0.005
School (Grade 10)-time mean	0.008	0.008	0.008
School (Grade 10)-time P5	0.007	0.008	0.008
School (Grade 10)-time P95	0.008	0.008	0.008
Fit Indexes			
Deviance mean	665,178	308,392	306,687
Deviance P5	664,875	308,224	306,458
Deviance P95	665,580	308,574	306,889
df	290,868	134,022	133,998
AIC mean	665,286	308,452	306,795
AIC P5	664,983	308,284	306,566
AIC P95	665,688	308,634	306,997
BIC mean	665,857	308,746	307,325
BIC P5	665,555	308,578	307,095
BIC P95	666,259	308,928	307,527

Note: ECE = early childhood education; SES = socioeconomic status; P5 = 5th percentile among imputations; P95 = 95th percentile among imputations; AIC = Akaike information criterion; BIC = Bayesian information criterion.